

COMP 472 Mini Project 2 - Short Reflection

Email Spam Classification AI System

Team Members

Name	Student ID
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Precise Contribution of Each Team Member

Noemie Corneillier: Dataset loading and validation, TF-IDF support, spam/ham visualization, and general code readability.

Kaoutar Machouat: Testing and debugging, checking edge cases, verifying confidence scores, testing the GUI/program flow, demo preparation, and the "having fun" experimentation part.

Yijia Xu: Model training, train/test split, classifier setup, accuracy, confusion matrix, confidence score logic, and prediction output.

Shared contribution: All three team members worked on understanding TF-IDF, since it was the main new AI concept needed to explain how text becomes numerical features.

Difficulties Faced and How We Overcame Them

- **Understanding TF-IDF** was the main challenge. We overcame it by reviewing examples and discussing how words are converted into numerical weights before training.
- **Splitting the work clearly** took planning. We assigned Kaoutar to testing/debugging and divided the remaining implementation work equally between Noemie and Yijia, while making sure everyone understood the full project.
- **Confidence scores and GUI testing** required extra checks. We tested many spam and ham examples, compared outputs, and adjusted the display so the result was easier to explain during the demo.

Actual Development Time Per Component

Our total time was close to the project estimate. The most important work was understanding TF-IDF, training/evaluating the model, and testing the full prediction flow.

Component	Time	Details
Understanding TF-IDF	2 hours	All members studied how TfidfVectorizer converts message text into n
Environment setup	1 hour	Set up Python/VS Code and installed pandas, numpy, scikit-learn, ma

Dataset loading	1 hour	Loaded the CSV with pandas and checked the required label,message
Feature extraction	1.5 hours	Applied TF-IDF to prepare the text data for machine learning.
Model training	2 hours	Trained the classifier using a train/test split.
Model evaluation	2 hours	Displayed accuracy and confusion matrix to measure performance.
Confidence scores	1 hour	Tested prediction probabilities and displayed confidence percentages.
Visualization	1 hour	Generated a chart showing the number of spam and ham messages.
Interactive loop / GUI	1.5 hours	Tested user input, quit behavior, prediction output, and GUI usability.
Testing and debugging	2.5 hours	Kaoutar tested the full program, edge cases, and demo examples.
Reflection/demo prep	1 hour	Prepared the reflection and reviewed the explanation for the demo.
Having fun	Unlimited	Experimented with different messages and possible improvements.

Conclusion

This project helped us understand how a supervised learning system can classify text as spam or ham. We practiced loading data, extracting TF-IDF features, training and evaluating a model, showing confidence scores, visualizing the dataset, and testing the system with new messages.